A Contrario Paradigm for YOLO-Based Infrared Small Target Detection

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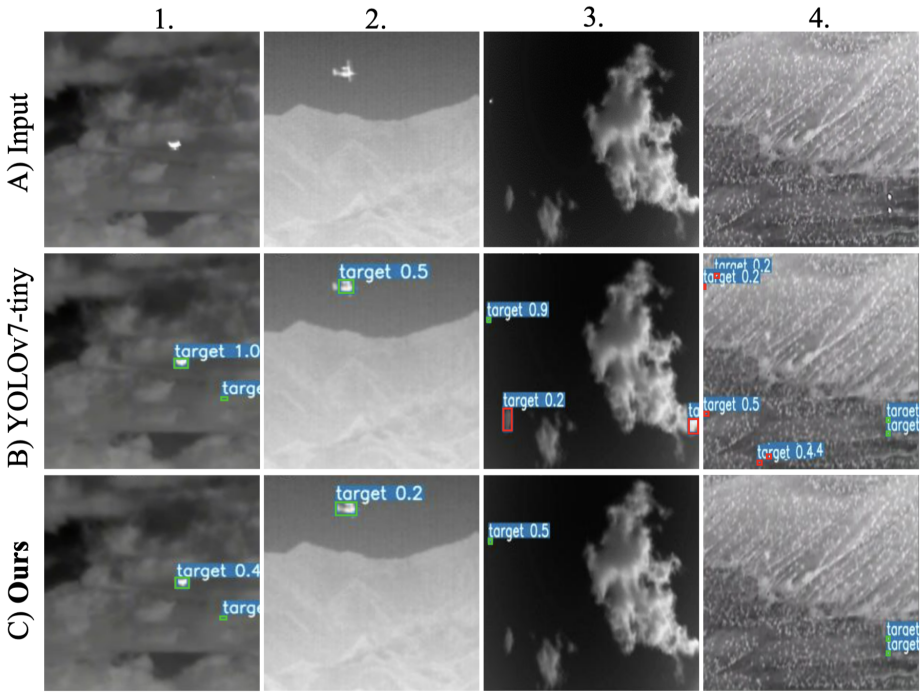
Outline

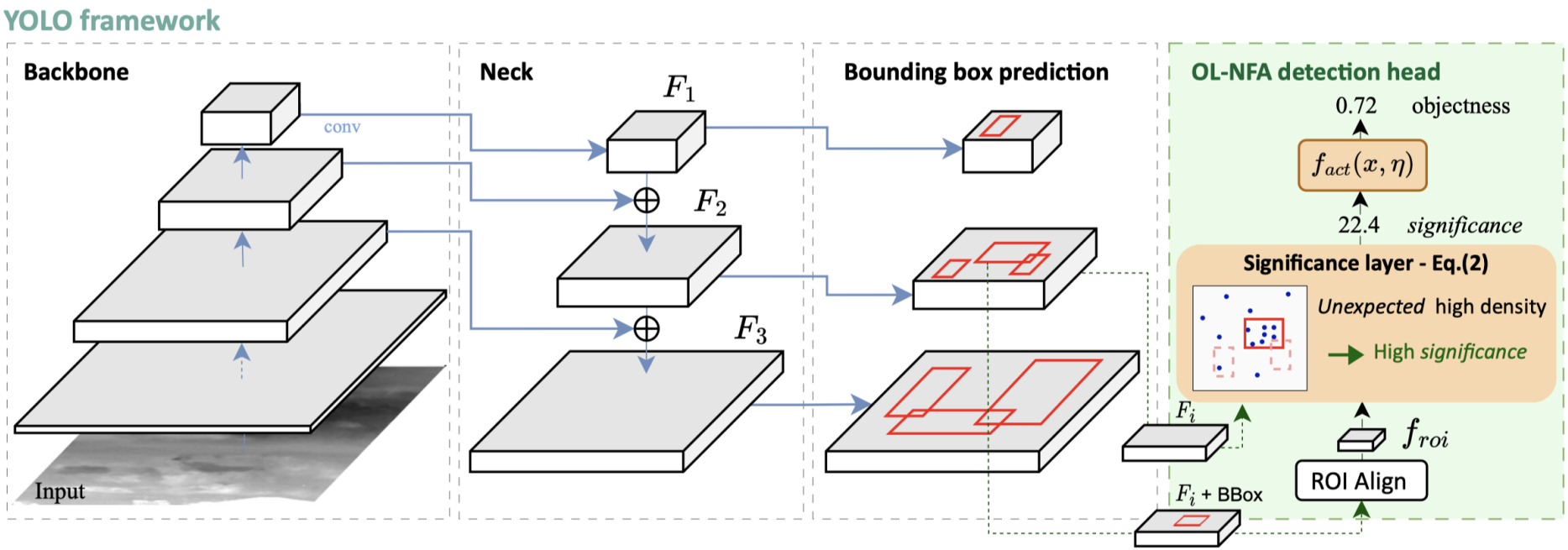
1. Overview
2. OL-NFA Principle
3. Implementation
4. Evaluation

Task: Infrared Small Target Detection

- Goal: detect very small objects in infrared (IR) images.
- Main difficulty:
  - targets often occupy only a few pixels,
  - backgrounds are noisy, and may look like targets leading to many false alarms.
- This makes IR small target detection a natural application for a **contrario / Number of False Alarms (NFA)** approach.

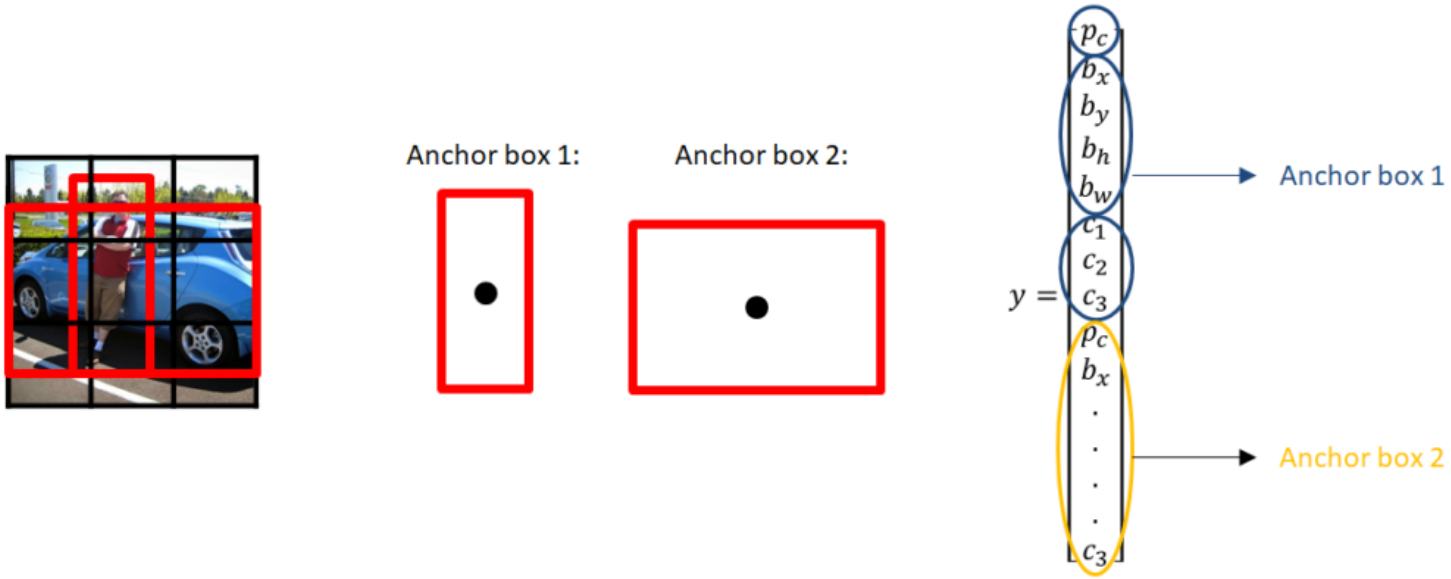


Overview (YOLO) – Feature Representation



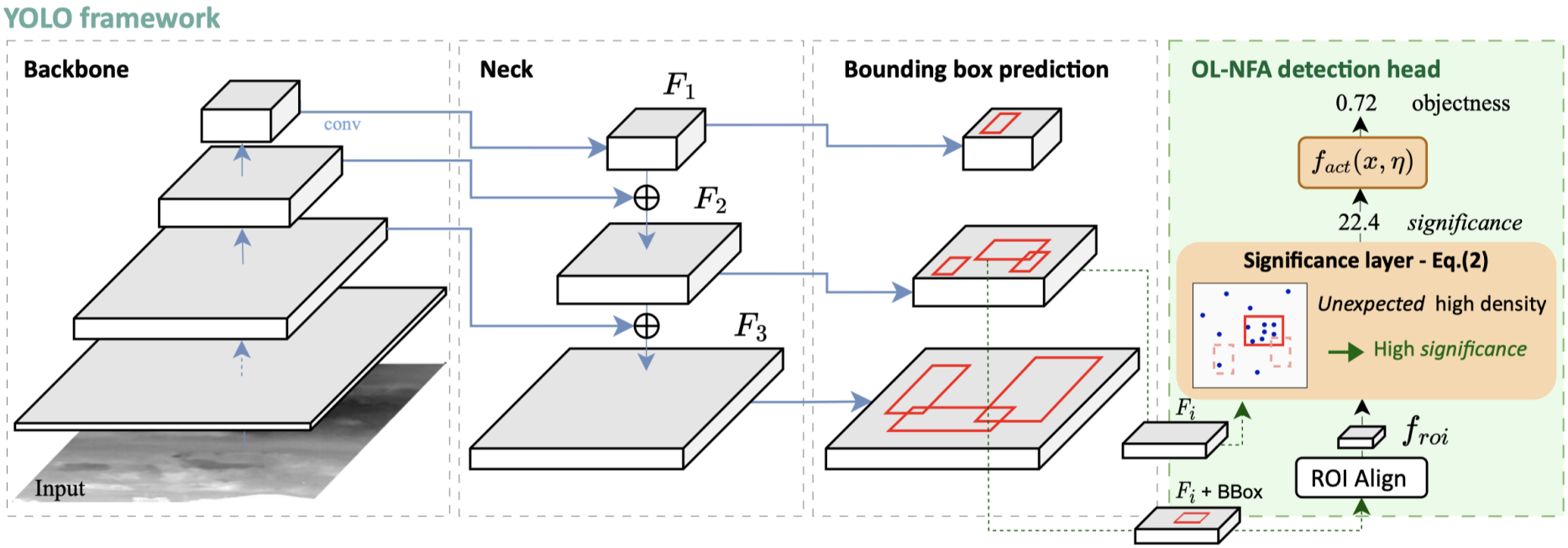
- Input image $(H \times W \times 1) \rightarrow$ feature maps F_i $(H_i \times W_i \times C_i)$
- The CNN progressively compresses spatial dimensions (height and width) while increasing the number of channels.
- Each feature map F_i captures high-level visual information at a different scale.
- Each spatial location (x_k, y_k) in F_i corresponds to a specific region of the original image.

Overview (YOLO) – Prediction Mechanism on F_i



- At each spatial location (x_k, y_k) of F_i, YOLO predicts multiple bounding boxes and scores (one per anchor).
- Bounding box parameters are predicted as offsets relative to predefined anchor boxes.

Overview (OL-NFA)



- The detector remains **YOLOv7-tiny**.
- Only the **objectness score** is re-estimated.
- So YOLO still decides *where* a possible object is.
- OL-NFA re-estimates *how statistically meaningful* this proposal is.

Naive Model and Family of Tests

- Naive model: the background is unstructured.
- On a binary map, each elementary position is active independently with probability p.
- For a tested region of size / volume ν, the number of active points follows:

$$K \mid H_0 \sim \mathrm{Binomial}(\nu, p).$$

- In this detector, the tested objects are not all rectangles in the image, they are the **candidate boxes proposed by YOLO**
- The Number of False Alarms is:

$$\mathrm{NFA}(\kappa, \nu, p) = \eta_{\text{test}} \sum_{t=\kappa}^{\nu} \binom{\nu}{t} p^t (1-p)^{\nu-t}$$

- We define **significance** of a detection as

$$S(B) = -\ln(\mathrm{NFA}(B)).$$

- Large significance means the detection is **unlikely under the background model**.

From Exact NFA to Differentiable Objectness

- For a fixed finite family of tests and discrete counts, the exact NFA has the classical guarantee:

$$\mathbb{E}\big[\#\{\text{detections with } \mathrm{NFA} \le \varepsilon\}\big] \le \varepsilon.$$

- However, in the implemented detector:
  - feature maps are real-valued, not binary,
  - the exact NFA is not convenient for backpropagation.
- So in practice, hard counts are replaced by soft counts:

$$\kappa(B) = \sum_{c,u,v} \sigma(R(B)_{cuv})$$

- The significance is approximated by **Hoeffding's inequality**:

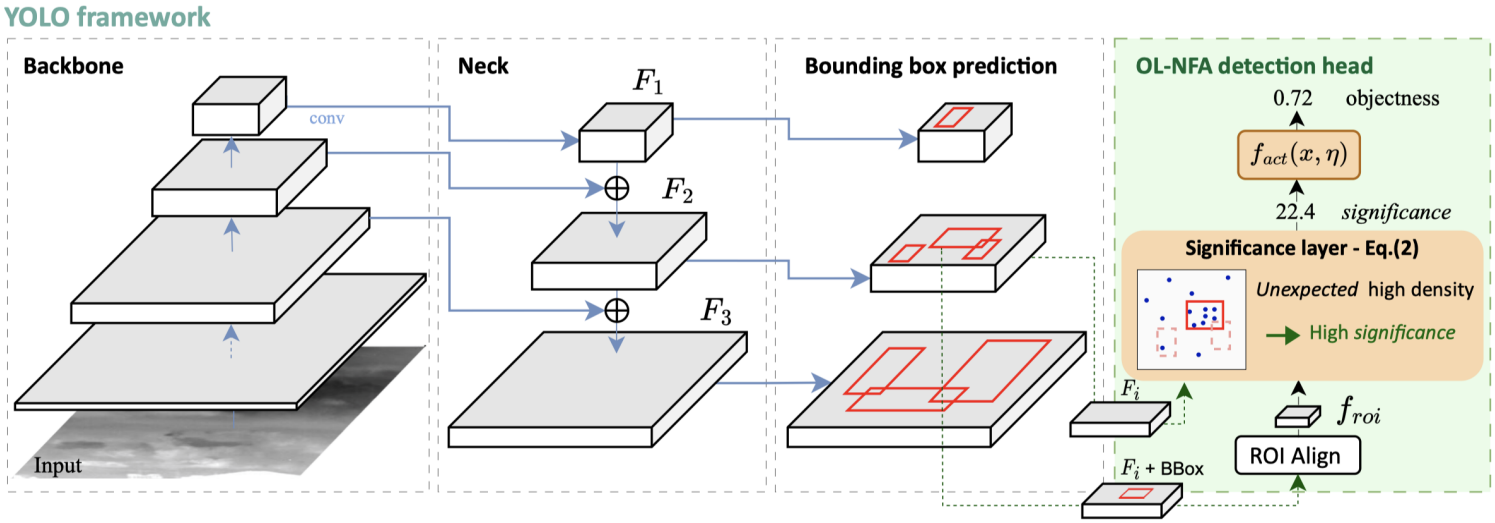
$$S(B) \approx \nu(B)\, D_{\mathrm{KL}}(q(B) \| p_i) - \ln \eta_{\text{test}}.$$

- This approximate significance is then converted into objectness:

$$o(B) = 2\,\sigma(S(B) + \ln \eta_{\text{test}}) - 1$$

- The detector can therefore be trained end-to-end like standard YOLO.

My Implementation



Key observation. The paper omits important details. The diagram is somewhat **misleading**, as it suggests applying the approach directly to F_i, while the method is actually described using a **2D NFA**

My approach. I keep the OL-NFA pipeline (YOLO proposals + ROI-based significance), but introduce:

- a **score-adapter convolution** before OL-NFA head.

Main Results on NUAA-SIRST

- **Post-processing:** NMS IoU threshold 0.05
- **Matching criterion:** prediction counted as correct if IoU ≥ 0.05

Table: Results on the full dataset.

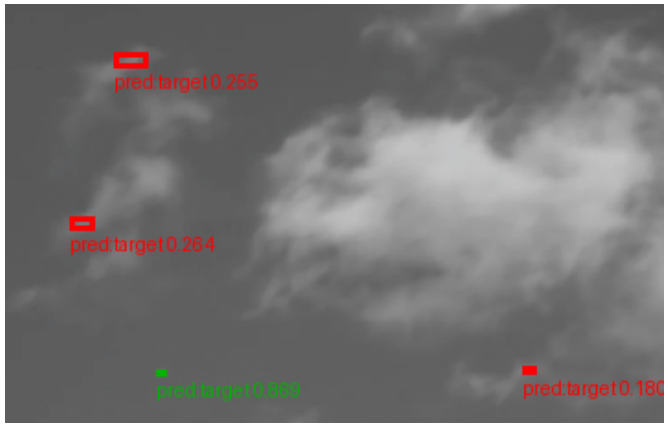
Run	F1	AP	Precision	Recall	FA/image
YOLOv7-tiny	**0.944**	**0.962**	**0.962**	**0.927**	**0.047**
YOLOv7-tiny + OL-NFA	0.928	0.918	**0.962**	0.897	**0.047**

Table: Few-shot results with 25 images.

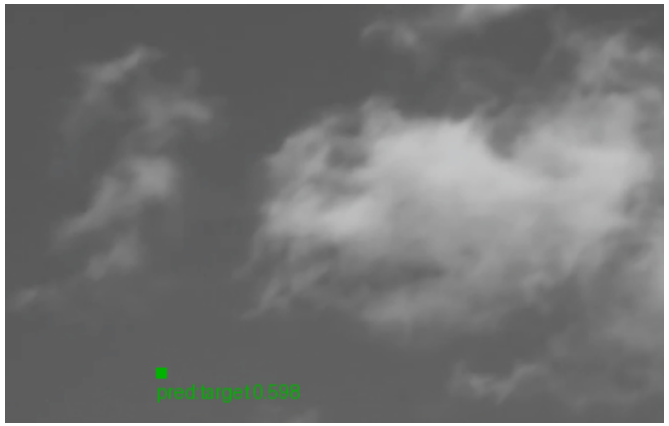
Run	F1	AP	Precision	Recall	FA/image
YOLOv7-tiny	**0.711**	**0.653**	0.795	**0.642**	0.209
YOLOv7-tiny + OL-NFA	0.663	0.595	**0.855**	0.541	**0.116**

- **Few-shot regime (25 images):**
  - Precision increases significantly: $0.795 \rightarrow 0.855$
  - False alarms are nearly halved: $0.209 \rightarrow 0.116$
  - Recall decreases

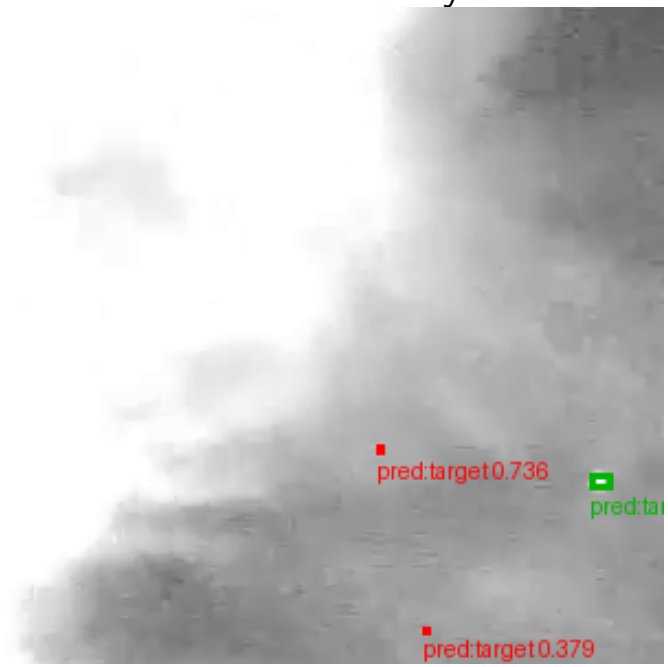
Qualitative Examples (Few-shot)



YOLOv7-tiny



YOLOv7-tiny + OL-NFA



YOLOv7-tiny

YOLOv7-tiny + OL-NFA

False positives in the baseline are removed with the OL-NFA module.

Conclusion

- Infrared small target detection is a difficult **false-alarm-sensitive** task.
- A **feature adapter** before the OL-NFA head Align improves performance by providing a consistent representation for NFA computation.
- Statistical meaningfulness can complement deep detectors.
- NFA module reduces false alarms and improves robustness in few-shot settings, at the cost of lower recall.
- **Future work:**
  - improve the **background probability estimation** p by making it **adaptive (spatial)** instead of global
  - design an NFA formulation that **directly exploits multi-channel features** instead of reducing them to scalar activations.

Bibliography

[1] A. Ciocarlan, S. Le Hegarat-Mascle, S. Lefebvre, A. Woiselle, and C. Barbanson. A Contrario Paradigm for YOLO-Based Infrared Small Target Detection. *arXiv preprint arXiv:2402.02288*, 2024.

[2] A. Ciocarlan. Thesis Small target detection using deep learning

[3] C.-Y. Wang, A. Bochkovskiy, and H.-Y. M. Liao. YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors. *arXiv preprint arXiv:2207.02696*, 2022.

[4] K. He, G. Gkioxari, P. Dollár, and R. Girshick. Mask R-CNN. *IEEE International Conference on Computer Vision (ICCV)*, 2017.

Appendix: Exact Link with Hoeffding / KL

From Homework (Hoeffding / Chernoff):

For a sum of independent variables $S_l = \sum_i X_i$ with mean p:

$$P\big(S_l \geq (p+t)l\big) \;\leq\; \exp\big(-\,l\,D_{\mathrm{KL}}(p+t\,\|\,p)\big).$$

Apply to detection:

- Random variable (under background model): $K = \sum_{i=1}^{\nu(B)} X_i$
- Observed value in box B: $\kappa(B)$
- Empirical activation:

$$q(B) = \frac{\kappa(B)}{\nu(B)}$$

Probability of a false alarm (Using Hoeffding bound):

$$P\big(K \geq \kappa(B)\big) = P\big(S_{\nu(B)} \geq q(B)\,\nu(B)\big) \;\leq\; \exp\big(-\,\nu(B)\,D_{\mathrm{KL}}(q(B)\|p)\big).$$

OL-NFA significance:

$$S(B) = -\log P \;\approx\; \nu(B)\,D_{\mathrm{KL}}(q(B)\|p).$$

Appendix: Author's thesis paper

Notation	Description
$\text{NFA}_{\mathcal{N}}$	Pixel-level NFA head based on Eq. (4.1).
$\text{NFA}_{\mathcal{U}_1}$	First version of the object-level NFA head introduced in Section 5.1.2. It is based on the 3D version of Eq. (5.1) where the third axis represents the channel-wise dimension of the feature maps.
$\text{NFA}_{\mathcal{U}_2}$	Second version of the object-level NFA head introduced in Section 5.1.3. It is also based on Eq. (5.1) but in this case the third axis represents a transformation of the feature map scores. The multi-channel aspect is taken into account by merging the NFA maps obtained independently for each channel (by taking the union of detections).

Method	F1	AP	Prec.	Rec.	#params
SOTA segmentation networks for IRSTD					
DNANet [8]	$\underline{96.9}^{\pm 0.5}$	$\underline{98.1}^{\pm 1.2}$	96.6	97.2	4.7 M
DNIM [8] + $\text{NFA}_{\mathcal{N}}$	$\mathbf{97.6}^{\pm 0.3}$	$\mathbf{98.4}^{\pm 0.6}$	97.9	97.4	4.8 M
Object detection methods					
YOLOv3 [48]	$96.1^{\pm 0.3}$	$97.5^{\pm 0.1}$	96.9	95.4	61.5 M
YOLOR [92]	$95.7^{\pm 2.2}$	$96.7^{\pm 1.1}$	96.5	94.9	52.5 M
YOLOv7 [49]	$96.5^{\pm 1.2}$	$97.6^{\pm 0.7}$	97.2	95.9	36.9 M
YOLOv7-tiny	$96.5^{\pm 0.6}$	$97.8^{\pm 0.4}$	96.9	96.2	6.0 M
YOLOv7-tiny + $\text{NFA}_{\mathcal{N}}$	$\mathbf{97.6}^{\pm 0.3}$	$\mathbf{98.3}^{\pm 0.1}$	97.6	97.6	6.5 M
YOLOv7-tiny + $\text{NFA}_{\mathcal{U}_1}$	$96.8^{\pm 1.6}$	$\underline{98.1}^{\pm 0.3}$	99.6	94.2	6.4 M
YOLOv7-tiny + $\text{NFA}_{\mathcal{U}_2}$	$\underline{97.0}^{\pm 0.3}$	$\mathbf{98.3}^{\pm 0.3}$	98.6	95.5	6.4 M

Appendix: Addition — Score Adapter on F_i

The figure in the paper is somewhat ambiguous, as it suggests applying ROI Align directly on the raw feature map F_i.
I therefore introduce a **trainable 4-layer convolutional adapter** per detection scale before ROI Align:

$$F_i \xrightarrow{\text{adapter}_i} \hat{F}_i \xrightarrow{\text{ROI Align}} f_{\text{roi}}$$

Layer	Operation	Output channels
1	Conv 1×1, BN, LeakyReLU	80
2	Conv 3×3, BN, LeakyReLU	80
3	Conv 3×3, BN, LeakyReLU	80
4	Conv 1×1	80

Applied at **3 scales** ($C_i = 128, 256, 512$): total $\approx$ **440** k parameters (consistent with the thesis's $\text{NFA}_{\mathcal{U}_1}$ overhead: 6.0 M $\rightarrow$ 6.4 M).

- **Fixed channel count for NFA.** The adapter projects all scales to $C = 80$ channels, providing a *consistent NFA volume* across scales.

Appendix: Dataset and Experimental Protocol

- **Dataset:** SIRST (427 infrared images, 1 class: `target`)
- **Split:** 256 / 85 / 86 for train / val / test
- **Models:**
  - YOLOv7-tiny
  - YOLOv7-tiny + OL-NFA
- **Training:** from scratch, 600 epochs, Adam, batch size 16
- **Settings:** input size 640×640, lr 10^{-3}
- **Post-processing:** NMS IoU threshold 0.05
- **Matching criterion:** prediction counted as correct if IoU ≥ 0.05
- **Metrics:** F1-score, AP, Precision, Recall

Appendix: What is ROI Align and Why Do We Need It?

The problem. To compute NFA significance of box B, we need the feature values of F_i *inside* B. But predicted boxes have arbitrary sizes, and F_i is spatially downsampled (stride 8, 16, or 32). We need a **fixed-size** feature tensor for any box, this is what ROI Align does.

ROI Align. Fix output size $r \times r$ (here $r = 5$).

1. **Project** box B onto F_i: divide corner coordinates by the stride. The box is now in feature-map space — possibly fractional and very small (e.g. 2.7×1.3 pixels for a tiny target).
2. **Divide** the projected box into an $r \times r$ grid of equal cells. Each cell covers $\frac{h_B}{r} \times \frac{w_B}{r}$ feature-map pixels. Different box sizes give different cell sizes, but always r^2 cells.
3. **Sample** inside each cell at $\texttt{sampling_ratio}^2 = 4$ evenly spaced sub-pixel locations.
4. **Interpolate** the feature value at each sub-pixel location by **bilinear interpolation** from the 4 surrounding integer-grid pixels Average the 4 values $\Rightarrow$ one scalar per cell per channel.

Output: $f_{\text{roi}} \in \mathbb{R}^{C \times r \times r}$, same size for every box regardless of its size or position.